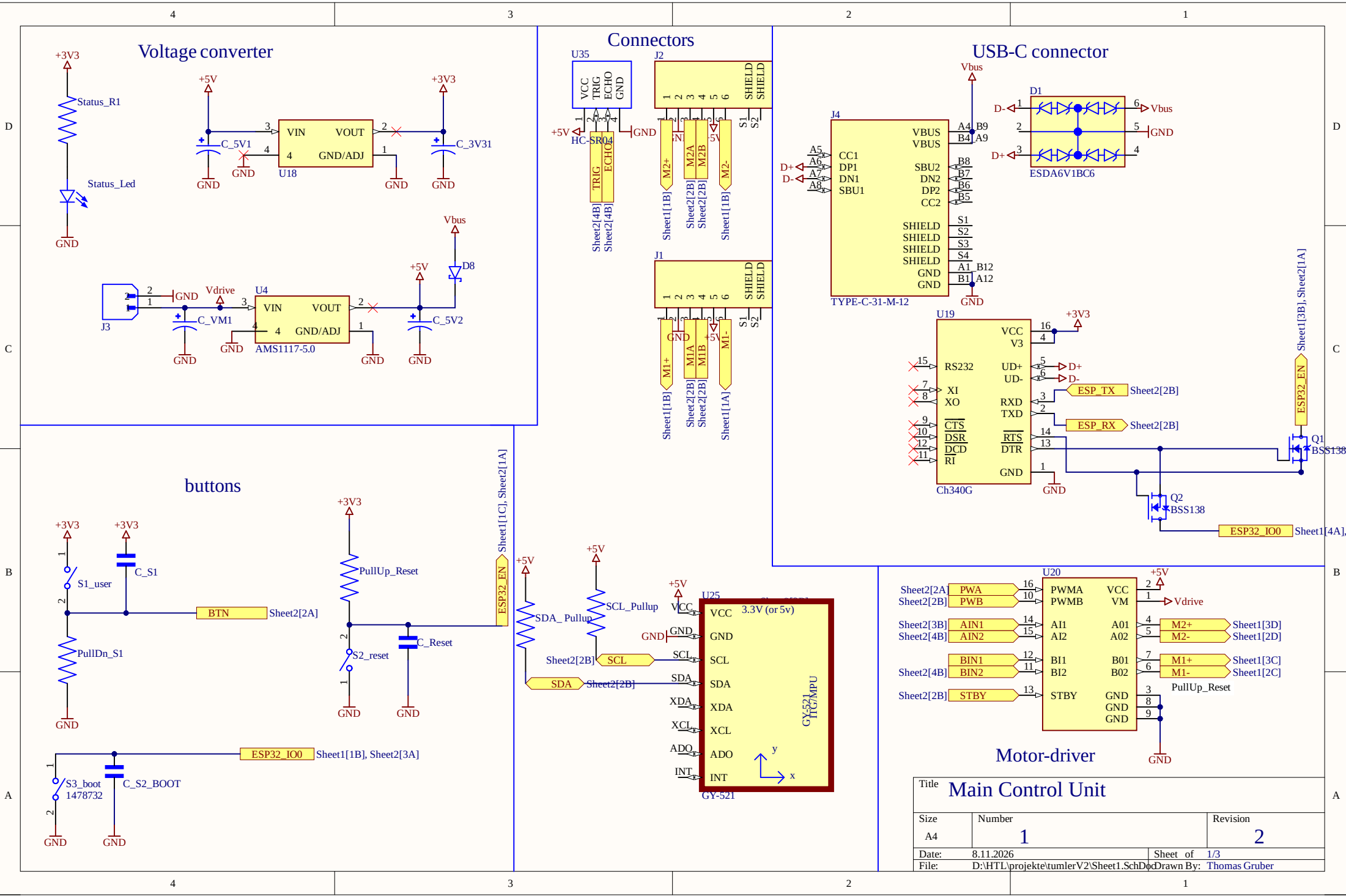
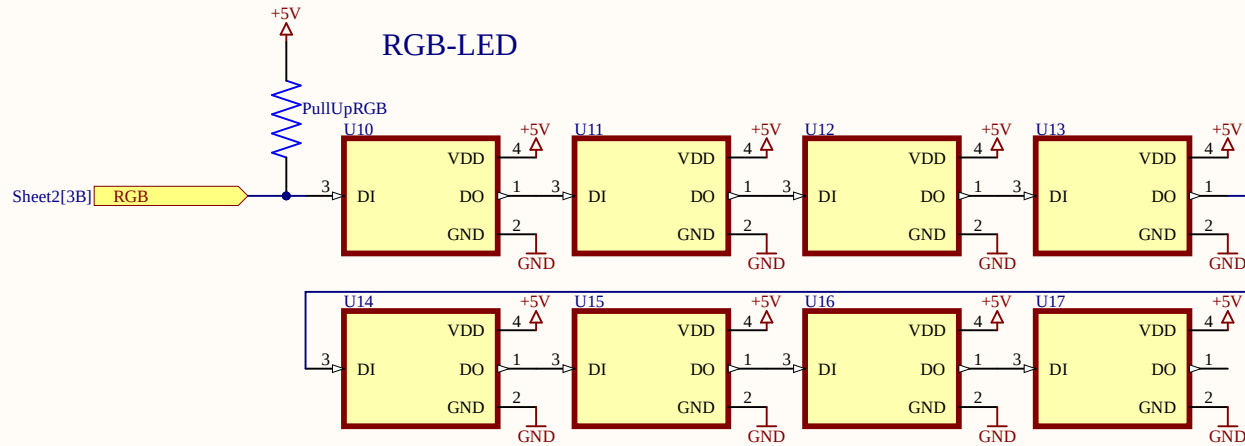


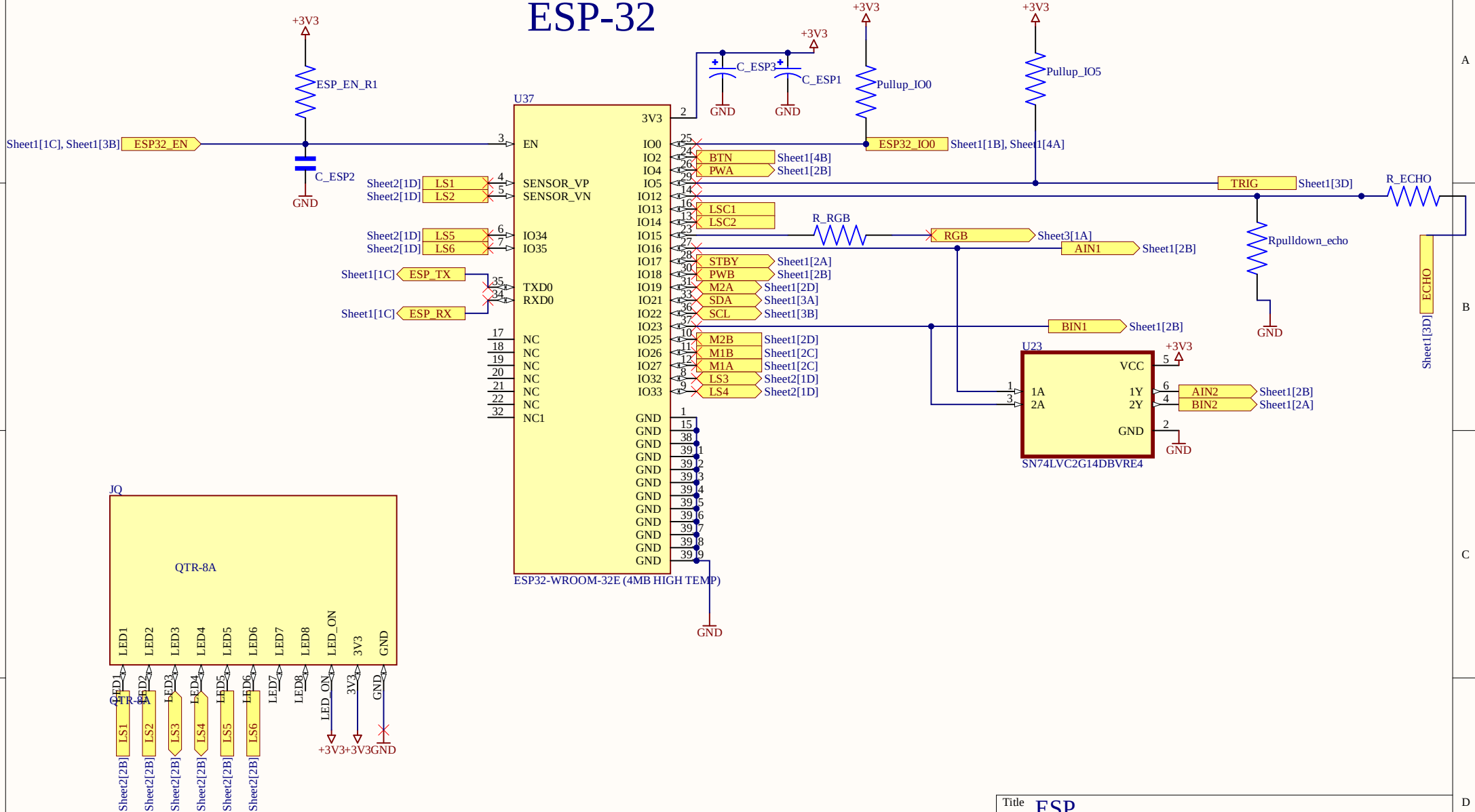


Title			Main Control Unit		
Size	Number			Revision	
A4	1			2	
Date:	8.11.2026			Sheet of	1/3
File:	D:\HTL\projekte\tumlerV2\Sheet1.SchDoc			Drawn By:	Thomas Gruber

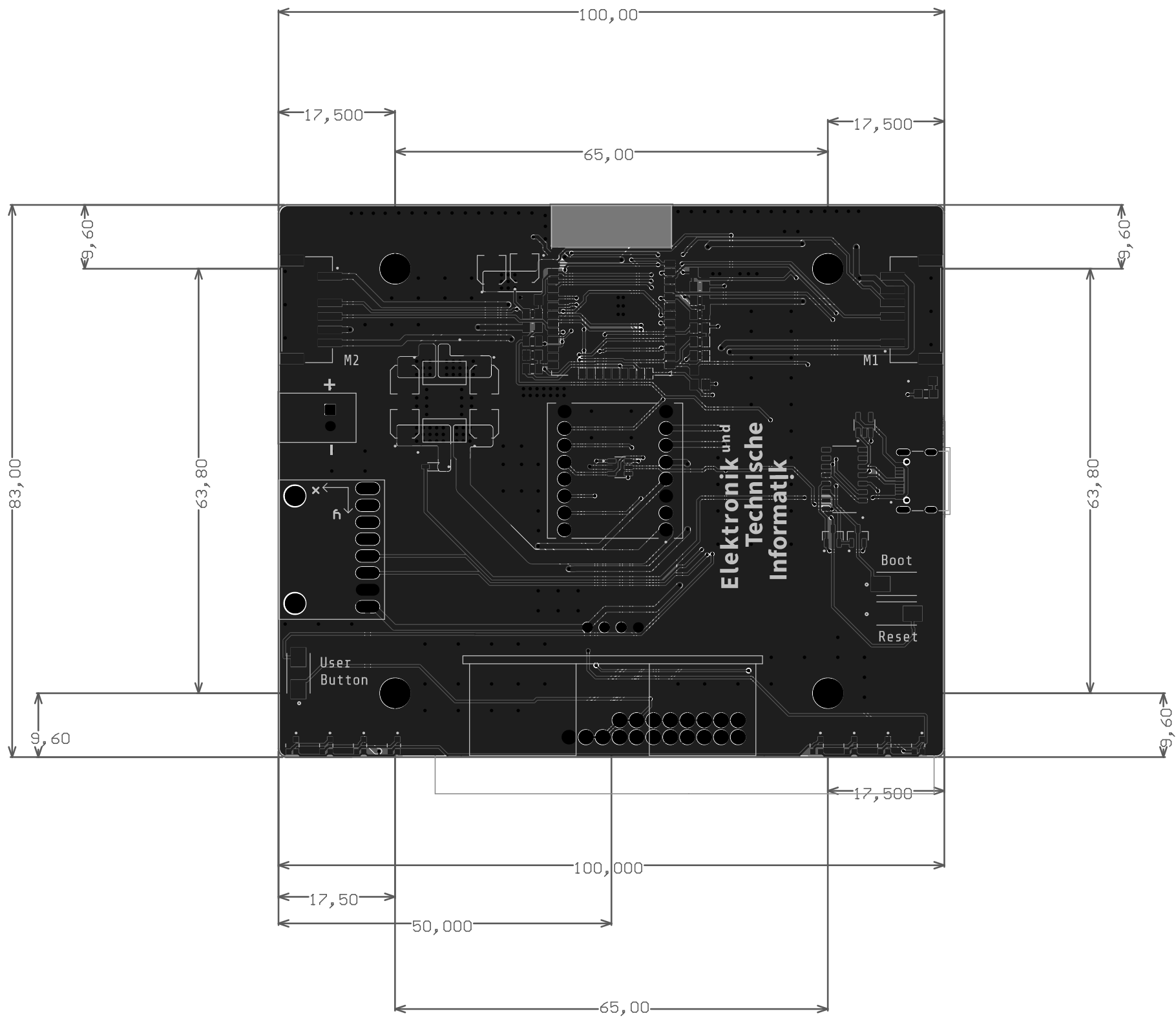


Title RGB-LEDs		
Size A4	Number 2	Revision 2
Date: 8.11.2026	Sheet of 2/3	
File: D:\HTL\projekte\tumlerV2\Sheet3.SchDoc	Drawn By: Thomas Gruber	

ESP-32



Title ESP		
Size A4	Number 3	Revision 2
Date: 8.11.2026	Sheet of 3/3	
File:	D:\HTL\projekte\tumlerV2\Sheet2.SchDoc Drawn By: Thomas Gruber	



Line #	Name	Description	Designator	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
1	EEEPK1E100R	10µF 25V Aluminum Electrolytic Capacitors Radial, Can - SMD 2000 Hrs @ 105°C	C_3V31, C_5V1, C_5V2, C_ESP3, C_VM1	5	Panasonic	EEEPK1E100R	Volume Production	JLC	C178538	0,05	0,25
	EEELHA010SR	Aluminum Electrolytic Capacitor, 2000h 4 (Dia) x 5.4mm, 1uF 50 VDC, S/V Series Panasonic Electronic Components EEELHA010SR	C_ESP1	1							
4	V0805A681KXCAP	CAP CER 100nF 200V COG/NPO 0805	C_ESP2, C_Reset, C_SL, C_S2_BOOT	4	Samsung Electro-Mechanics	CL21B104KCFNNINE	Manual Solution	JLC	C28233	0,0057	0,0228
6	ESDA6V1B06	- Clamp - Ipp Tvs Diode Surface Mount SOT-23-6	D1	1	STMicroelectronics	ESDA6V1B06	Volume Production	JLC	C495220	0,1549	0,1549
7	B5819WS-TP	Diode Schottky 40 V 1A Surface Mount SOD 323	D8	1	Guangdong Hottech	1N5819WS	Manual Solution	JLC	C191023	0,0095	0,0095
8	0805W8F1002T5E	10K Ohm 1/8W Thick Film Resistor 0805 Chip - Surface Mount	ESP_EN_R1, PullDn_SL, PullUp_I00, PullUp_I05, PullUp_Reset, PullUpRGB, Rpulldown_echo, SQ_Pullup, SDA_Pullup	9	Royal Ohm	0805W8F1002T5E					
9	SBB-PH-SM4-TB (LP)(SN)	Connector Header Surface Mount, Right Angle 6 position 0.079 (2.00mm)	J1, J2	2	JST	SBB-PH-SM4-TB(LP)(SN)	Volume Production	JLC	C265405	0,528	1,06
10	S2B-XHA-1(LP)(SN)	Connector Header Through Hole, Right Angle 2 position 0.098 (2.50mm)	J3	1	CJT(Changjiang Connectors)	A2501WRB-2P	Manual Solution	JLC	C273216	0,0344	0,0344
11	TYPEC-31-M-12	USB Connectors 24 Receptacle 1 8.94* 7.3mm RoHS	J4	1	Korean Hroparts Etec	TYPEC-31-M-12	Manual Solution	JLC	C165948	0,1697	0,1697
12	QTR-8A	QRE1113GRseries Phototransistor Sensor Evaluation Board	JQ	1							
13	BSSL38	N-Channel 50 V 220mA(Ta) 350mW (Ta) Surface Mount SOT-23-3	Q1, Q2	2	onsemi	BSSL38	Volume Production	JLC	C52895	0,0464	0,0928
	CROV0805470RFFR0EC	470 Ohms ±1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) - Thick Film	R_ECHO, R_RGB	2							
	1478732	Switch, Switch, Tact, Smt, J Lead, 4.3Mm TE Connectivity 147873-2	SL_user, S2_reset, S3_boot	3							
15	APHOM2012SURCK-F01	Red 630nm LED Indication - Discrete 1.95V 0805 (2012 Metric)	Status_Led	1	Hubei KBVTO Etec	KT-0805Y	Manual Solution	JLC	C2296	0,122	0,122
16	ERA6AH-D200V		Status_R1	1	UNI-ROYAL(Uniroyal Etec)	0805W8F2000T5E	Manual Solution	JLC	C17540	0,0018	0,0018
18	AMSL117-5.0		U4	1	Advanced Monolithic Systems	AMSL117-5.0	Manual Solution	JLC	C6187	0,1913	0,1913
17	WS2812-2020	Intelligent control LED light source, SMD-4	U10, U11, U12, U13, U14, U15, U16, U17	8	Worldsemi	WS2812B-2020	Manual Solution	JLC	C965555	0,08	0,64
19	AMSL117-3.3	AMSL117-3v3	U18	1	Advanced Monolithic Systems	AMSL117-3.3	Manual Solution	JLC	C6186	0,1653	0,1653
20	Ch340G	USB Interface Integrated Circuit	U19	1	WCH(Jiangsu Qm Heng)	CH340G	Manual Solution	JLC	C14267	0,5025	0,5025
21	ROB-14450	TB6612FNG Motor Controller/Driver Power Management Evaluation Board	U20	1							
22	SN74LVC2G14DBVRE4	Dual Schmitt-Trigger Inverter 6-SOT-23 -40 to 125	U23	1	Texas Instruments	SN74LVC2G14DBVRE4	Volume Production	JLC	C12401	0,0681	0,0681
23	GY-521	Triple Axis Accelerometer Gyro Breakout	U25	1							
24	HCSR04	Ultrasonic Sensor Module	U35	1							
25	ESP32-WROOM-32E (4MB HIGH TEMP)	ESP32-WROOM-32E 4IN	U37	1	Espressif Systems	ESP32-WROOM-32-N4	Manual Solution	JLC	C82899	4,12	4,12